

# HELIX

## Use Case Textual Descriptions

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# Introduction

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This document provides textual descriptions for all use cases modelled in the **HELIX Platform Use Case Diagram**. Each description follows the standard academic template: *Name*, *Actors*, *Preconditions*, *Postconditions*, *Main Scenario*, and *Alternative Scenarios*. Sub-flow use cases («include» / «extend») are grouped with their parent for conciseness.

## Actors

| ID | Actor                | Description                                                                       |
|----|----------------------|-----------------------------------------------------------------------------------|
| A0 | Authenticated User   | Abstract generalisation; shared capabilities inherited by all authenticated roles |
| A1 | Administrator        | Platform and user management authority                                            |
| A2 | Blue Team Operator   | SOC analyst; monitors events, manages incidents                                   |
| A3 | Red Team Operator    | Penetration tester; executes offensive tools                                      |
| A4 | Purple Team Operator | Senior hybrid engineer; inherits Blue + Red capabilities                          |
| A5 | Visitor              | Unauthenticated user; browses public content, registers, logs in                  |
| A6 | Learner              | Authenticated student; studies, tracks progress, takes notes                      |
| A7 | Detection Engine     | Autonomous system actor; runs the detection pipeline                              |

## Use Case Groups

1. Access Control (UC-01 to UC-06)
2. Security Operations (UC-07 to UC-10)
3. Detection Pipeline (UC-11 to UC-16)
4. Red Team Orchestration (UC-17 to UC-20)

5. Collaboration & Knowledge Base (UC-21 to UC-26)
6. Learner Workspace (UC-27 to UC-30)
7. AI Assistance (UC-31 to UC-33)

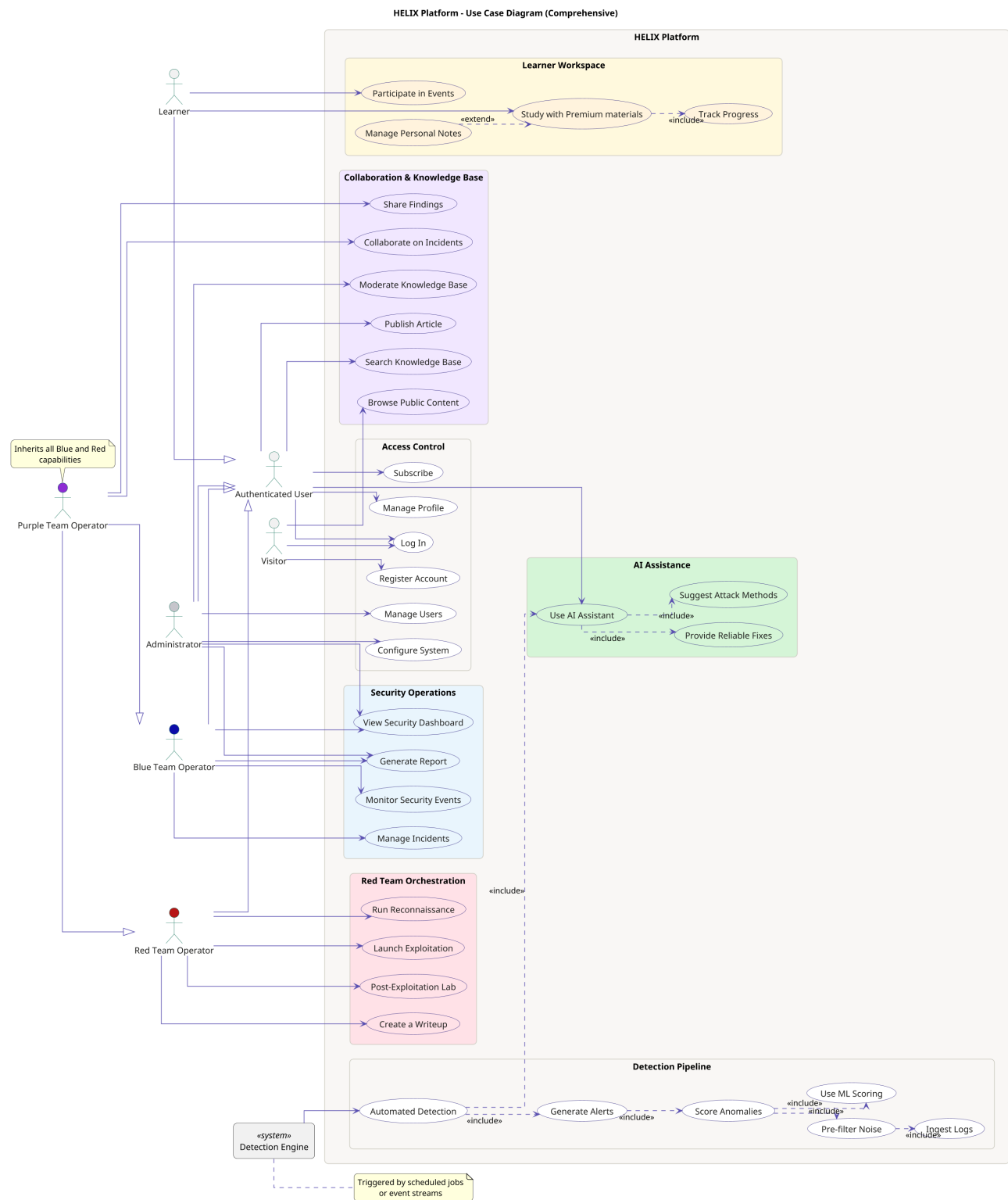


Figure 1: HELIX Platform – Use Case Diagram (Comprehensive)

**Figure Notes:** Actor generalisation arrows show inheritance (e.g., Purple Team Operator inherits from both Blue Team Operator and Red Team Operator). Use case relationships use «include» and «extend» stereotypes. The Detection Engine is a system actor outside the system boundary.

# 1. Access Control

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## 1.1 UC-01 – Register Account

### UC-01 Register Account

**Actors:** Visitor (primary)

**Preconditions:** User has no existing HELIX account.

**Postconditions:** A new Learner account is created; user can log in.

**Main Scenario:**

1. Visitor opens the registration form.
2. Visitor submits username, email, and password.
3. System validates input (unique username, password strength).
4. System hashes the password (bcrypt) and persists the account.
5. System confirms registration success.

**Alternatives:**

- [Alt-A] Username already exists → error displayed.
- [Alt-B] Password too weak → inline validation message.

## 1.2 UC-02 – Log In

### UC-02 Log In

**Actors:** Visitor (initiates); Authenticated User (inherits)

**Preconditions:** User has a registered, active account.

**Postconditions:** A valid session is created; user redirected to role dashboard.

**Main Scenario:**

1. User submits username and password.
2. System validates credentials against stored bcrypt hash.
3. System creates a PHP session with role information.
4. User is redirected to the role-appropriate dashboard.

**Alternatives:**

- **[Alt-A]** Invalid credentials → error message displayed.
- **[Alt-B]** Account deactivated → access denied.

### 1.3 UC-03 – Manage Profile

#### UC-03 Manage Profile

**Actors:** Authenticated User (all concrete roles inherit)

**Preconditions:** User is logged in with a valid session.

**Postconditions:** Profile information updated and persisted.

**Main Scenario:**

1. User navigates to profile settings.
2. User modifies fields (display name, email, password).
3. System validates and applies changes.

**Alternatives:**

- **[Alt-A]** New email already in use → validation error.
- **[Alt-B]** Password mismatch → update blocked.

### 1.4 UC-04 – Manage Users

#### UC-04 Manage Users

**Actors:** Administrator (primary)

**Preconditions:** Admin is authenticated.

**Postconditions:** User account created, deactivated, or role-reassigned.

**Main Scenario:**

1. Admin opens the user management panel.
2. Admin selects an action: create, edit role, or deactivate.
3. System applies the change and logs it to the audit trail.

**Alternatives:**

- **[Alt-A]** Admin tries to deactivate themselves → rejected.
- **[Alt-B]** Target user is logged in → session invalidated.



## 1.5 UC-05 – Configure System

### UC-05 Configure System

**Actors:** Administrator (primary)

**Preconditions:** Admin is authenticated.

**Postconditions:** System parameters (detection thresholds, API keys, retention) updated.

**Main Scenario:**

1. Admin navigates to system configuration.
2. Admin modifies parameters.
3. System validates and saves; affected services reload.

**Alternatives:**

- [Alt-A] Invalid value → validation error.
- [Alt-B] API key incorrect → flagged on next use.

## 1.6 UC-06 – Subscribe

### UC-06 Subscribe

**Actors:** Authenticated User (primary)

**Preconditions:** User is logged in. A subscription plan is available.

**Postconditions:** User subscription tier upgraded; premium features unlocked.

**Main Scenario:**

1. User navigates to the subscription page.
2. User selects a plan and confirms.
3. System records the subscription and updates feature access.

**Alternatives:**

- [Alt-A] Already subscribed → current plan shown.

## 2. Security Operations

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### 2.1 UC-07 – View Security Dashboard

#### UC-07 View Security Dashboard

**Actors:** Administrator, Blue Team Operator, Red Team Operator, Purple Team Operator, Learner

**Preconditions:** User is authenticated. At least one security event exists.

**Postconditions:** Dashboard displayed with live metrics; no state modified.

**Main Scenario:**

1. User navigates to the Security Dashboard.
2. System fetches metrics: active alerts by severity, 24-hour timeline, open incidents.
3. Dashboard renders role-appropriate content.

**Alternatives:**

- **[Alt-A]** No events exist → empty state with guidance displayed.
- **[Alt-B]** Learner view → learning-oriented metrics shown.

### 2.2 UC-08 – Monitor Security Events

#### UC-08 Monitor Security Events

**Actors:** Blue Team Operator (primary), Purple Team Operator

**Preconditions:** User is authenticated. Log events are available.

**Postconditions:** Filtered list of security events displayed.

**Main Scenario:**

1. Operator opens the event monitoring view.
2. Operator applies filters: severity, date range, source IP, event type.
3. System returns matching events with enrichment data.
4. Operator reviews details and may escalate to an incident.

**Alternatives:**

- **[Alt-A]** No events match → empty result; operator adjusts criteria.

## 2.3 UC-09 – Manage Incidents

### UC-09 Manage Incidents

**Actors:** Blue Team Operator (primary), Purple Team Operator, Administrator

**Preconditions:** User is authenticated. At least one alert exists.

**Postconditions:** Incident created or updated; status transition logged.

**Main Scenario:**

1. Operator selects alerts and creates an incident.
2. System assigns status `Open` with creation timestamp.
3. Operator transitions status: `Open` → `Investigating` → `Resolved`, adding notes.
4. Each change recorded in the incident timeline.

**Alternatives:**

- **[Alt-A]** Invalid transition → rejected; allowed transitions shown.

## 2.4 UC-10 – Generate Report

### UC-10 Generate Report

**Actors:** Blue Team Operator (primary), Administrator

**Preconditions:** User is authenticated. At least one incident or alert exists.

**Postconditions:** Structured PDF report generated and available for download.

**Main Scenario:**

1. User selects scope: single incident, date range, or summary.
2. System compiles data: timeline, severity distribution, recommendations.
3. System generates a PDF and provides a download link.

**Alternatives:**

- **[Alt-A]** No data in scope → warning; prompt to adjust scope.

## 3. Detection Pipeline

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The Detection Engine pipeline is a chain of «include» relationships: UC-11 (Automated Detection) includes UC-15 (Generate Alerts) → UC-14 (Score Anomalies) → UC-13 (Pre-filter Noise) → UC-12 (Ingest Logs). UC-14 also includes UC-16 (Use ML Scoring). UC-11 includes UC-31 (Use AI Assistant).

### 3.1 UC-11 – Automated Detection

#### UC-11 Automated Detection

**Actors:** Detection Engine (primary system actor)

**Preconditions:** Detection Engine is running. Log sources are reachable.

**Postconditions:** Alerts generated for anomalous events; pipeline completed.

**Main Scenario:**

1. Detection Engine triggered by scheduler or event stream.
2. «include» UC-12 – Ingest Logs.
3. «include» UC-13 – Pre-filter Noise.
4. «include» UC-14 – Score Anomalies (includes UC-16).
5. «include» UC-15 – Generate Alerts for HIGH/CRITICAL.
6. «include» UC-31 – Use AI Assistant for enrichment.
7. System notifies assigned analysts.

**Alternatives:**

- [Alt-A] Log source unreachable → error logged; pipeline continues.
- [Alt-B] ML model unavailable → rule-based detection continues.
- [Alt-C] AI service offline → alerts generated without AI enrichment.

## 3.2 UC-12 – Ingest Logs

### UC-12 Ingest Logs (included by UC-13)

**Actors:** Detection Engine

**Preconditions:** Log source accessible; parser registered.

**Postconditions:** Raw logs parsed and normalised as structured entries.

**Main Scenario:**

1. Engine reads raw logs (Syslog, JSON, Apache, Windows Event Log).
2. Parser selects appropriate format handler.
3. Each entry normalised into structured fields and written to staging store.

**Alternatives:**

- [Alt-A] Unknown format → tagged as unparseable; flagged for review.

## 3.3 UC-13 – Pre-filter Noise

### UC-13 Pre-filter Noise (included by UC-14)

**Actors:** Detection Engine

**Preconditions:** Normalised log entries in staging store.

**Postconditions:** Low-signal events suppressed; high-signal events forwarded.

**Main Scenario:**

1. Noise filter evaluates each log entry.
2. Duplicate or benign patterns discarded; suppression reason logged.
3. Remaining events forwarded to scoring stage.

**Alternatives:**

- [Alt-A] Filter unavailable → all events pass through.

## 3.4 UC-14 – Score Anomalies

### UC-14 Score Anomalies (included by UC-15)

**Actors:** Detection Engine

**Preconditions:** Filtered events available. ML model loaded.

**Postconditions:** Each event receives severity: LOW, MEDIUM, HIGH, or CRITICAL.

**Main Scenario:**

1. «include» UC-16 – Use ML Scoring (Isolation Forest).
2. Anomaly score mapped to severity per thresholds.

3. Scored events written to detection store.

**Alternatives:**

- [Alt-A] ML model fails → rule-based fallback assigns severity.

### 3.5 UC-15 – Generate Alerts

**UC-15 Generate Alerts (included by UC-11)**

**Actors:** Detection Engine

**Preconditions:** UC-14 completed; HIGH or CRITICAL events exist.

**Postconditions:** Alert records created; analysts notified; audit log written.

**Main Scenario:**

1. System selects events scored HIGH or CRITICAL.
2. For each, an alert record created with timestamp, source IP, severity.
3. System enriches alert with GeoIP and reputation data.
4. Notifications dispatched to Blue Team Operators.

**Alternatives:**

- [Alt-A] Enrichment APIs unreachable → alert created without enrichment.

### 3.6 UC-16 – Use ML Scoring

**UC-16 Use ML Scoring (included by UC-14)**

**Actors:** Detection Engine

**Preconditions:** Isolation Forest model loaded and up to date.

**Postconditions:** Each event receives a continuous anomaly score.

**Main Scenario:**

1. Engine constructs feature vector per event (event frequency, error rate, time-of-day, source IP diversity).
2. Isolation Forest returns anomaly score.
3. Score forwarded to UC-14 for severity mapping.

**Alternatives:**

- [Alt-A] Feature extraction fails → event scored MEDIUM by default.

## 4. Red Team Orchestration

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### 4.1 UC-17 – Run Reconnaissance

#### UC-17 Run Reconnaissance

**Actors:** Red Team Operator (primary), Purple Team Operator

**Preconditions:** User authenticated with Red Team role. Legal consent accepted. Target in scope.

**Postconditions:** Reconnaissance results persisted and viewable.

**Main Scenario:**

1. Operator selects tool (nmap, WHOIS, OSINT).
2. Operator specifies target and scan parameters.
3. System executes tool server-side; results streamed back.
4. Results normalised and stored in scan history.

**Alternatives:**

- **[Alt-A]** Tool timeout → partial results stored; error flagged.
- **[Alt-B]** Target out of scope → execution blocked; attempt logged.

### 4.2 UC-18 – Launch Exploitation

#### UC-18 Launch Exploitation

**Actors:** Red Team Operator (primary), Purple Team Operator

**Preconditions:** Reconnaissance complete. Legal consent accepted.

**Postconditions:** Exploitation result logged (success or failure).

**Main Scenario:**

1. Operator selects a vulnerability from recon findings.
2. Operator configures an exploit module.
3. System executes exploit and captures result.
4. Result (success/failure, CVE reference) persisted.

**Alternatives:**

- **[Alt-A]** Exploit fails → failure logged; alternative suggested.

- **[Alt-B]** Consent not accepted → features blocked.

## 4.3 UC-19 – Post-Exploitation Lab

### UC-19 Post-Exploitation Lab

**Actors:** Red Team Operator (primary), Purple Team Operator

**Preconditions:** Successful exploitation or lab environment available.

**Postconditions:** Post-exploitation actions logged for analysis.

**Main Scenario:**

1. Operator enters the post-exploitation lab environment.
2. Operator performs actions: privilege escalation, lateral movement, exfiltration simulation.
3. Each action recorded for blue team awareness.

**Alternatives:**

- **[Alt-A]** Lab environment unavailable → retry after environment reset.

## 4.4 UC-20 – Create a Writeup

### UC-20 Create a Writeup

**Actors:** Red Team Operator (primary), Purple Team Operator

**Preconditions:** User authenticated. An operation or CTF result exists.

**Postconditions:** Writeup submitted for moderation; visible upon approval.

**Main Scenario:**

1. Operator opens the writeup editor.
2. Operator drafts content, attaches screenshots and scan results.
3. Operator submits; system tags and queues for Admin moderation.

**Alternatives:**

- **[Alt-A]** Writeup rejected → operator notified with reason; can revise.



## 5. Collaboration & Knowledge Base

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### 5.1 UC-21 – Browse Public Content

#### UC-21 Browse Public Content

**Actors:** Visitor (primary)

**Preconditions:** None. User is unauthenticated.

**Postconditions:** Public content displayed; no session created.

**Main Scenario:**

1. Visitor navigates to the HELIX public homepage.
2. System renders publicly available articles, writeups, and platform features.
3. Visitor browses content without logging in.

**Alternatives:**

- **[Alt-A]** Visitor accesses restricted page → redirected to login.

### 5.2 UC-22 – Search Knowledge Base

#### UC-22 Search Knowledge Base

**Actors:** Authenticated User (all concrete roles inherit)

**Preconditions:** User is logged in. Published articles exist.

**Postconditions:** Matching articles displayed; no state changed.

**Main Scenario:**

1. User enters a search query (keyword, tag, or difficulty).
2. System performs full-text search and returns ranked results.
3. User selects and reads an article.

**Alternatives:**

- **[Alt-A]** No results → system suggests related tags.

### 5.3 UC-23 – Publish Article

#### UC-23 Publish Article

**Actors:** Authenticated User (all concrete roles inherit)

**Preconditions:** User is logged in.

**Postconditions:** Article submitted; visible after Admin moderation.

**Main Scenario:**

1. User opens the article editor.
2. User writes content, adds tags (topic, team, difficulty).
3. User submits; article enters **Pending** status, queued for moderation.
4. On approval, article becomes searchable (UC-22).

**Alternatives:**

- **[Alt-A]** Article flagged → user notified; draft preserved.

### 5.4 UC-24 – Collaborate on Incidents

#### UC-24 Collaborate on Incidents

**Actors:** Purple Team Operator (primary)

**Preconditions:** User authenticated as Purple Team. Active incident exists.

**Postconditions:** Incident updated with cross-team annotations; timeline entry created.

**Main Scenario:**

1. Operator opens an active incident.
2. Operator adds cross-team annotations: correlating Red findings with Blue alerts.
3. System records annotation with author and timestamp.

**Alternatives:**

- **[Alt-A]** Incident closed → view-only; can reopen with justification.

### 5.5 UC-25 – Share Findings

#### UC-25 Share Findings

**Actors:** Purple Team Operator (primary)

**Preconditions:** User authenticated as Purple Team. Completed operation or incident exists.

**Postconditions:** Findings shared; notification sent.

**Main Scenario:**

1. Operator selects findings (scan results, timeline, writeup) to share.

2. Operator designates recipients (team or individual).
3. System packages findings and notifies recipients in-app.

**Alternatives:**

- **[Alt-A]** Recipient has insufficient clearance → sharing blocked.

## 5.6 UC-26 – Moderate Knowledge Base

**UC-26   Moderate Knowledge Base**

**Actors:** Administrator (primary)

**Preconditions:** Admin authenticated. Articles pending review exist.

**Postconditions:** Article approved (published), rejected, or archived.

**Main Scenario:**

1. Admin opens the moderation queue.
2. Admin reviews each pending article for guideline compliance.
3. Admin approves → article goes live; or rejects with reason → author notified.

**Alternatives:**

- **[Alt-A]** Admin archives published article → removed from search, preserved.
- **[Alt-B]** Admin deletes article → requires confirmation.

## 6. Learner Workspace

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### 6.1 UC-27 – Study with Premium Materials

#### UC-27 Study with Premium Materials

**Actors:** Learner (primary)

**Preconditions:** Learner authenticated with active subscription. Premium content exists.

**Postconditions:** Premium module accessed; progress updated (UC-29 included).

**Main Scenario:**

1. Learner navigates to the premium learning section.
2. Learner selects a module (video, lab, quiz).
3. System renders content and records access.
4. «include» UC-29 – Track Progress: completion updated.

**Alternatives:**

- [Alt-A] Subscription expired → access blocked; redirect to subscribe.
- [Alt-B] Learner wants notes → «extend» UC-28.

### 6.2 UC-28 – Manage Personal Notes

#### UC-28 Manage Personal Notes («extend» UC-27)

**Actors:** Learner (primary)

**Preconditions:** Learner authenticated. A module or article is open.

**Postconditions:** Note created, updated, or deleted; data strictly private.

**Main Scenario:**

1. Learner opens the Notes panel alongside a module or article.
2. Learner creates, edits, or deletes a note.
3. System persists the note, linked to the content; visible only to owner and Admin.

**Alternatives:**

- [Alt-A] Learner searches notes by keyword → matching notes displayed.

## 6.3 UC-29 – Track Progress

### UC-29 Track Progress (included by UC-27)

**Actors:** Learner (indirect, via UC-27)

**Preconditions:** Learner completes or accesses a module/exercise.

**Postconditions:** Progress log updated; dashboard refreshed.

**Main Scenario:**

1. System records completed module, quiz score, or CTF flag in progress log.
2. Learner's dashboard updated with new completion percentage.

**Alternatives:**

- [Alt-A] Module already completed → marked as revisited; no duplicate entry.

## 6.4 UC-30 – Participate in Events

### UC-30 Participate in Events

**Actors:** Learner (primary)

**Preconditions:** Learner authenticated. An active event (CTF, webinar, challenge) is live.

**Postconditions:** Learner registered for or completed the event; results logged.

**Main Scenario:**

1. Learner browses the events calendar.
2. Learner registers for an upcoming event or joins a live event.
3. Learner submits flags or answers; results recorded on the leaderboard.
4. Event completion logged in the progress tracker (via UC-29).

**Alternatives:**

- [Alt-A] Event at capacity → learner added to waitlist.
- [Alt-B] Deadline passed → registration closed; archived materials viewable.

## 7. AI Assistance

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### 7.1 UC-31 – Use AI Assistant

#### UC-31 Use AI Assistant

**Actors:** Authenticated User (all concrete roles inherit)

**Preconditions:** User is logged in. OpenAI API available. Alert or query context exists.

**Postconditions:** AI-generated explanation, remediation, or attack suggestion returned.

**Main Scenario:**

1. User opens the AI Assistant panel (from alert, incident, or freeform chat).
2. System constructs a prompt with alert context and relevant log data.
3. «include» UC-32 – Provide Reliable Fixes.
4. «include» UC-33 – Suggest Attack Methods.
5. AI response displayed; conversation history persisted per session.

**Alternatives:**

- [Alt-A] OpenAI API unavailable → degraded mode message displayed.
- [Alt-B] User submits CVE identifier → CVE summary returned.

### 7.2 UC-32 – Provide Reliable Fixes

#### UC-32 Provide Reliable Fixes (included by UC-31)

**Actors:** AI Service (internal)

**Preconditions:** UC-31 initiated; alert context available.

**Postconditions:** Context-specific remediation steps returned.

**Main Scenario:**

1. OpenAI API receives alert context and queries knowledge base for matching patterns.
2. System returns prioritised remediation checklist (patch, block IP, rotate credentials).

**Alternatives:**

- [Alt-A] No known mitigation → generic hardening guidance returned.

## 7.3 UC-33 – Suggest Attack Methods

### UC-33 Suggest Attack Methods (included by UC-31)

**Actors:** AI Service (internal)

**Preconditions:** UC-31 initiated; Red Team or Purple Team context detected.

**Postconditions:** Relevant attack techniques returned.

**Main Scenario:**

1. OpenAI API identifies the attack surface or target technology from context.
2. System returns relevant attack techniques, tools, and exploitation paths.

**Alternatives:**

- **[Alt-A]** Legal consent not accepted → output blocked with consent prompt.

## Conclusion

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This document provides complete textual descriptions for all **33 use cases** (UC-01 to UC-33) of the HELIX platform, fully aligned with the SRS and Use Case Diagram. All descriptions follow the academic format: *Name, Actors, Preconditions, Postconditions, Main Scenario, and Alternative Scenarios*. The eight actors (Authenticated User, Administrator, Blue Team Operator, Red Team Operator, Purple Team Operator, Visitor, Learner, Detection Engine) are consistently applied.